ABRAHAM ISRAEL ROCKLYN

AI Engineer | Machine Learning & Cloud Engineer | Full-Stack Web & WordPress Developer

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SUMMARY

AI Engineer who builds **LLM applications, retrieval-augmented generation (RAG) and multi-agent systems** end to end: data and models, APIs and web interfaces, and the **AWS and Kubernetes** infrastructure they run on. At Rivet Health, built LangGraph agent workflows with human approval gates and RAG pipelines with source-grounded citations; before that, researched LLM reliability and evaluation at Texas A&M and supported AWS data center operations. Also a **full-stack web and WordPress developer** who ships fast, search-friendly sites. Every system ships with automated tests, CI and a measured evaluation; **23 open-source projects** document the design and results. Pursuing an **M.S. in Computer Science (AI)**.

TECHNICAL SKILLS

AI & LLMs: LLM applications, RAG, AI agents & multi-agent systems (LangGraph), tool calling, prompt engineering, embeddings, vector search (pgvector), hybrid search & reranking, LLM evaluation, guardrails & prompt-injection defense, Claude API

Machine Learning: scikit-learn, XGBoost, SHAP, ONNX Runtime, MLflow, feature engineering, model evaluation, forecasting, recommender systems, anomaly detection, A/B testing, OCR, speech-to-text (Whisper)

Languages & Backend: Python, Go, TypeScript, JavaScript, PHP, SQL; FastAPI, Flask, REST APIs, PostgreSQL, Apache Kafka, asynchronous processing

Cloud & DevOps: AWS (EC2, S3, IAM, VPC, RDS, Lambda, Step Functions, DynamoDB, EventBridge, SQS, SageMaker, Glue, Athena), Terraform, Docker, Kubernetes, KEDA, GitHub Actions CI/CD, Prometheus, Grafana, Linux

Frontend & Practices: React, TypeScript, interactive dashboards; system design, automated testing, observability, secure API design, access control, audit logging, AI-assisted development (Claude, ChatGPT)

Web & WordPress: WordPress (Elementor, Gutenberg, Divi, WPBakery, ACF), WooCommerce, Shopify, PHP, HTML5, CSS3, JavaScript, responsive & accessible design (WCAG 2.1/2.2), Core Web Vitals, SEO (Yoast, Rank Math, Ahrefs, SEMrush), GA4, Google Tag Manager, cPanel, DNS & SSL

PROFESSIONAL EXPERIENCE

AI Engineer | Rivet Health 2024 – 2026

- Built **multi-agent workflows in LangGraph** with tool execution, persisted workflow state, **human approval gates**, retries, recovery and audit trails, so automated steps can be paused, reviewed and safely resumed.
- Engineered **RAG pipelines**: document ingestion, OCR, embeddings, **hybrid keyword + vector retrieval** and reranking, producing answers grounded in and cited to their sources.
- Built **evaluation frameworks and automated tests** measuring answer quality, retrieval accuracy, citation correctness, workflow reliability and security.
- Designed and developed **full-stack AI applications** from concept to tested release: architecture, backend services, **REST APIs**, database schemas, **React front ends** and model integrations.
- Delivered cloud-ready architectures with **Docker, Terraform, AWS** and **GitHub Actions CI/CD**, and used Claude and ChatGPT to speed research, implementation, debugging, testing and documentation.

Artificial Intelligence Research Assistant | Texas A&M Engineering Experiment Station 2023 – 2024

- Researched applied **LLMs, AI agents and semantic retrieval** with a focus on reliability: hallucination prevention, source attribution, access control and responsible AI.
- Designed **controlled evaluation datasets and experiments** to measure retrieval quality, answer accuracy, model behavior and system performance.
- Prototyped integrations across external model APIs, **local ONNX models**, vector databases and distributed inference services; documented architecture decisions, trade-offs and limitations.

Software Engineering Assistant | Southwestern University 2022 – 2023

- Built a **full-stack web application (Python, Flask, React, PostgreSQL)** that asynchronously analyzes GitHub interactions, with API endpoints for retrieving and visualizing participation.
- Designed and implemented **REST APIs** to store and manage data from learning management systems.

Data Center Operations Technician | Amazon Web Services (AWS) 2020 – 2022

- Diagnosed and resolved hardware and infrastructure incidents in **AWS data centers**; installed, maintained and replaced server components to keep capacity available.
- Worked with engineering and operations teams on incident investigation, following security procedures and maintaining technical documentation and incident records.

SELECTED PROJECTS

Open source with tests, CI and published evaluations at israelrocklyn.info. *Results are measured on documented synthetic datasets and benchmarks.*

Autonomous Business Operations Platform | *Python, LangGraph, Claude API, FastAPI, PostgreSQL, React*

- Multi-agent platform automating refunds, supplier invoices and purchase requests, with persistent checkpoints, **human-in-the-loop approvals**, permission-guarded tools, idempotent actions, a tamper-evident audit trail and per-run token budgets.
- **96.9% correct outcomes** on 129 labeled cases, **100% approval precision and recall**, and **zero unauthorized money movement** in security tests.

Intelligent Document AI | *Python, FastAPI, OCR, pgvector, ONNX Runtime, React*

- Document platform for OCR, classification, field extraction, hybrid search and question answering with **page-level citations** and document-level access control.
- **Recall@5 of 1.0** with entity-constrained retrieval, **0.968 correct-page citations**, 100% field extraction and a 1.6% OCR character error rate.

LLM Guardrails Gateway | *Python, scikit-learn, FastAPI, Docker, Claude API*

- Security gateway in front of any LLM: **prompt-injection detection, PII redaction**, output policy and spend budgets.
- **0% attack success** across 3,680 test prompts (a keyword filter let 90.9% through) at **3.1 ms p95** added latency.

Distributed AI Inference Platform | *Go, Python, ONNX Runtime, Kubernetes, KEDA, Prometheus, Grafana*

- Go gateway and Python model workers with tenant quotas, **admission control**, deadline propagation, **dynamic batching**, KEDA autoscaling and graceful draining.
- At 2x overload served **38.6 req/s vs 0.05** without admission control; kept **98.7%** of interactive requests flowing during a batch flood; **zero errors** when a replica was killed under load.

AI Research & Knowledge Engine | *Python, FastAPI, pgvector, ONNX cross-encoder, React*

- Team knowledge bases with hybrid search, cross-encoder reranking, source-conflict detection and workspace access control: **100% Recall@5, 99.3% citation precision** and 100% correct "not found" answers.

AI Developer Intelligence | *Python, tree-sitter, pgvector, ONNX, FastAPI, React*

- Codebase intelligence across 5 languages: semantic code search, cited answers with file and line, call graphs, security findings and change-impact analysis: **95% search Recall@5** and **100% precision and recall** on 21 planted findings.

Fraud Detection Engine | *Python, Apache Kafka, XGBoost, SHAP, FastAPI, React*

- Real-time fraud scoring on a Kafka stream with SHAP explanations and an analyst case dashboard: **93% of fraud dollars caught** at about 31 alerts a day, 96% precision in the critical band.

israelrocklyn.info: Portfolio Website | *PHP, JavaScript, HTML5, CSS3, GitHub API*

- Custom PHP site with no framework: 23 project case studies, SEO structured data, a CSRF-protected contact form, a strict Content Security Policy and a **private analytics dashboard** (bcrypt login, CSRF, throttling) charting visits and GitHub traffic.

Also: LLM Evaluation Platform (calibrated judge, κ 0.91 vs humans), Multimodal AI Assistant (voice, image and PDF; 92% grounded accuracy), AWS Serverless Event Platform (0 violations under 566 injected faults), AWS MLOps Platform, Demand Forecasting, Experimentation Platform, Feature Store, Recommender System, AWS Resilience Lab.

EDUCATION

Master of Science in Computer Science, Concentration in Artificial Intelligence Expected Jan 2028
Southern New Hampshire University (SNHU)

Bachelor of Science (Honors) in Information Systems
Institute of Public Administration and Management (IPAM)